# Svolgimento TE del 18/07/2024

## Gestione Path

In[∘]:=
```
SetDirectory[NotebookDirectory[]]
```
Out[∘]=
```
C:\Scrittura\Temi Esame TeX\18_07_2024 - Robotica\Svolgimento
```

## Caricamento package iDynTree

Caricamento

In[∘]:=
```
<< iDynTree`;
```

Funzioni disponibili

In[∘]:=
```
? iDynTree`*
```

## Risposta ai vari quesiti

In[∘]:=
```
? RevoluteTwist
```
Out[∘]=

Symbol

RevoluteTwist[q, w] builds the 6–vector

corresponding to point q on the axis with unit vector w for a revolute joint.

In[∘]:=
```
? PrismaticTwist
```
Out[∘]=

Symbol

PrismaticTwist[q, w] builds the 6–vector

corresponding to point q on the axis with unit vector w for a prismatic joint.

### Q1: calcolo espressioni esplicite degli $[\xi_i]_S$ e $M$ (anche per ispezione!)

In[∘]:=
```
ξ1s = {0, 0, 0, 0, 0, 1}
```
Out[∘]=
```
{0, 0, 0, 0, 0, 1}
```

In[∘]:=
```
ξ2s = {0, -L1, 0, 0, 0, 1}
```
Out[∘]=
```
{0, -L1, 0, 0, 0, 1}
```

In[•]:=
```
ξ3s = {0, -(L1 + L2), 0, 0, 0, 1}
```
Out[•]=
```
{0, -L1 - L2, 0, 0, 0, 1}
```

In[•]:=
```
ξ4s = {0, -(L1 + L2 + L3), 0, 0, 0, 1}
```
Out[•]=
```
{0, -L1 - L2 - L3, 0, 0, 0, 1}
```

In[•]:=
```
? SpatialJacobianGlobalPOE
```
Out[•]=

Symbol

SpatialJacobianGlobalPOE[{xi1,th1}, ... ,{xiN, thN} ,g0] computes

the Spatial Manipulator Jacobian of a robot defined by the given twists

described in the Global POE form.

In[•]:=
```
M = RPToHomogeneous[{
    {1, 0, 0},
    {0, 1, 0},
    {0, 0, 1}
   }, {L1 + L2 + L3 + L4, 0, 0}];
M // MatrixForm
```
Out[•]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 & L1+L2+L3+L4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

In[•]:=
```
Jsbs[θ1_, θ2_, θ3_, θ4_] = SpatialJacobianGlobalPOE[
    {{ξ1s, θ1}, {ξ2s, θ2}, {ξ3s, θ3}, {ξ4s, θ4}, M] // FullSimplify;
Jsbs[θ1, θ2, θ3, θ4] // MatrixForm
```
Out[•]//MatrixForm=

$$\begin{pmatrix} 0 & L1\,Sin[\theta1] & L1\,Sin[\theta1]+L2\,Sin[\theta1+\theta2] & L1\,Sin[\theta1]+L2\,Sin[\theta1+\theta2]+L3\,Sin[\theta1+\theta2+\theta \\ 0 & -L1\,Cos[\theta1] & -L1\,Cos[\theta1]-L2\,Cos[\theta1+\theta2] & -L1\,Cos[\theta1]-L2\,Cos[\theta1+\theta2]-L3\,Cos[\theta1+\theta2+ \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

In[•]:=
```
Jsbb[θ1_, θ2_, θ3_, θ4_] =
  BodyJacobianGlobalPOE[{{ξ1s, θ1}, {ξ2s, θ2}, {ξ3s, θ3}, {ξ4s, θ4}, M] // FullSimplify;
Jsbb[θ1, θ2, θ3, θ4] // MatrixForm
```
Out[•]//MatrixForm=

$$\begin{pmatrix} L3\,Sin[\theta4]+L2\,Sin[\theta3+\theta4]+L1\,Sin[\theta2+\theta3+\theta4] & L3\,Sin[\theta4]+L2\,Sin[\theta3+\theta4] & L3\,S \\ L4+L3\,Cos[\theta4]+L2\,Cos[\theta3+\theta4]+L1\,Cos[\theta2+\theta3+\theta4] & L4+L3\,Cos[\theta4]+L2\,Cos[\theta3+\theta4] & L4+L3 \\ 0 & 0 & \\ 0 & 0 & \\ 0 & 0 & \\ 1 & 1 & \end{pmatrix}$$

In[•]:=
```
JsbsConfig = Jsbs[0, 0, Pi / 2, -Pi / 2]; JsbsConfig // MatrixForm
```
Out[•]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 0 & L3 \\ 0 & -L1 & -L1-L2 & -L1-L2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

In[∘]:= 
```
ws1 = {10, 10, 0, 0, 0, 10}
```
Out[∘]= $\{10, 10, 0, 0, 0, 10\}$

In[∘]:= 
```
Transpose[JsbsConfig].ws1 // FullSimplify
```
Out[∘]= $\{10, -10\,(-1 + L1), -10\,(-1 + L1 + L2), -10\,(-1 + L1 + L2 - L3)\}$

In[∘]:= 
```
ws2 = {-10, 10, 0, 0, 0, -10}
```
Out[∘]= $\{-10, 10, 0, 0, 0, -10\}$

In[∘]:= 
```
Transpose[JsbsConfig].ws2 // FullSimplify
```
Out[∘]= $\{-10, -10\,(1 + L1), -10\,(1 + L1 + L2), -10\,(1 + L1 + L2 + L3)\}$

In[∘]:= 
```
? ForwardKinematics
```

Symbol

Out[∘]= ForwardKinematics[{xi1,th1}, ... ,{xiN,thN}, g0]

computes the forward kinematics via the product of exponentials formula.

g0 is the initial affine tranformation if any.